

CBSE Class 12 Physics — Half-Yearly Predicted Paper (Set 8 of 10)

Section A: 1-Mark Questions (MCQ/Assertion-Reason)

Q1. A point charge q is placed exactly at one of the corners of a cube. The total electric flux passing through the entire cube is: (a) q/ϵ_0 (b) $q/8\epsilon_0$ (c) $q/6\epsilon_0$ (d) $q/4\epsilon_0$ [Practice] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q2. A solid conducting sphere of radius R is given a total charge Q . The electric potential and the electric field strictly at the geometric centre of the sphere are respectively: (a) Zero and Zero (b) $Q/4\pi\epsilon_0 R$ and Zero (c) $Q/4\pi\epsilon_0 R$ and $Q/4\pi\epsilon_0 R^2$ (d) Zero and $Q/4\pi\epsilon_0 R^2$ [PYQ 2021] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q3. The relaxation time (τ) of free electrons in a metallic conductor: (a) increases strictly with an increase in temperature. (b) decreases strictly with an increase in temperature. (c) does not depend on the temperature of the conductor. (d) changes unpredictably with temperature. [PYQ 2020] | **[Chapter 3: Current Electricity | Confidence: High]**

Q4. A proton moves with a velocity v perpendicularly to a uniform magnetic field B , describing a circular path of radius r . If the velocity of the proton is suddenly doubled ($2v$) in the same magnetic field, the new radius of its circular path will be: (a) r (b) $2r$ (c) $r/2$ (d) $4r$ [PYQ 2019] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q5. For which of the following classes of magnetic materials is the magnetic permeability (μ) maximum? (a) Diamagnetic (b) Paramagnetic (c) Ferromagnetic (d) Non-magnetic [PYQ 2022] | **[Chapter 5: Magnetism & Matter | Confidence: Very High]**

Q6. According to Faraday's laws of electromagnetic induction, the magnitude of the induced emf in a circuit is directly proportional to the: (a) magnetic flux linked with the circuit. (b) rate of change of magnetic flux linked with the circuit. (c) electrical resistance of the circuit. (d) area of the closed circuit. [PYQ 2023] | **[Chapter 6: Electromagnetic Induction | Confidence: Very High]**

Q7. In a pure inductive AC circuit, if the frequency of the applied AC source is exactly doubled, the current in the circuit will: (a) double (b) become exactly half (c) remain the same (d) become four times [Practice] | **[Chapter 7: Alternating Current | Confidence: High]**

Q8. In a plane electromagnetic wave, the amplitude of the oscillating electric field is 60 V/m . The corresponding amplitude of the oscillating magnetic field will be: (a) $2 \times 10^7 \text{ T}$ (b) $2 \times 10^{-7} \text{ T}$ (c) $6 \times 10^{-7} \text{ T}$ (d) $1.8 \times 10^{10} \text{ T}$ [PYQ 2020] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q9. A thin symmetric convex lens of focal length f is precisely cut into two identical halves along its principal axis. The focal length of each individual part will physically be: (a) f (b) $f/2$ (c) $2f$ (d) $4f$ [PYQ 2021] | **[Chapter 9: Ray Optics | Confidence: High]**

Q10. In a Young's double-slit experiment, if the monochromatic source slit is shifted slightly upwards (parallel to the plane of the slits), the central bright fringe on the screen will: (a) shift upwards (b) shift downwards (c) completely vanish (d) strictly remain at the center [Practice] | **[Chapter 10: Wave Optics | Confidence: High]**

Q11. According to Einstein's photoelectric equation, the maximum kinetic energy of the emitted photoelectrons completely depends on: (a) the intensity of the incident light only. (b) the frequency of the incident light and the nature of the material. (c) the physical distance of the light source from the metal. (d) the surface area of the illuminated metal. [PYQ 2020] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Q12. Which of the following fundamental particles, if all are moving with the exact same velocity, will possess the longest associated de-Broglie wavelength? (a) Electron (b) Proton (c) Alpha particle (d) Neutron [PYQ 2019] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

For Questions 13 to 16, two statements are given — one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes (a), (b), (c), and (d) as given below. (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is NOT the correct explanation of A. (c) A is true but R is false. (d) A is false and R is also false.

Q13. Assertion (A): The net electric field anywhere inside a hollow, statically charged spherical conductor is always identically zero. **Reason (R):** Under electrostatic equilibrium, excess charges strictly reside entirely on the outer surface of a conductor. [PYQ 2022] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q14. Assertion (A): The equivalent electrical resistance of a parallel combination of multiple resistors is strictly less than the smallest individual resistance in the combination. **Reason (R):** In a parallel circuit, the total current divides among the parallel branches while the potential difference strictly remains constant across each branch. [Practice] | **[Chapter 3: Current Electricity | Confidence: High]**

Q15. Assertion (A): The magnetic field produced by a long, straight current-carrying wire is inversely proportional to the perpendicular distance from the wire. **Reason (R):** According to the Biot-Savart law, the magnetic field due to any small current element is inversely proportional to the distance from the element. [PYQ 2018] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q16. Assertion (A): A convex lens made of glass (focal length f) behaves as a diverging (concave) lens when it is completely immersed in a transparent liquid whose refractive index is greater than that of the glass. **Reason (R):** The focal length of a lens becomes strictly negative when it is placed in a surrounding medium of a higher refractive index. [PYQ 2017] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Section B: 2-Mark Questions (Very Short Answer)

Q17. A 12 pF parallel plate capacitor is connected across a 50 V battery. Calculate the total electrostatic energy permanently stored in the capacitor. [PYQ 2017] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q18. Define the term 'drift velocity' of charge carriers in a conductor. Write the fundamental mathematical relation connecting the drift velocity to the macroscopic electric current flowing through the conductor. [PYQ 2014, 2020] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q19. An aircraft having a wingspan of 40 m flies directly horizontally eastward with a constant speed of 1080 km/h at a constant altitude in the northern hemisphere. What is the motional emf induced strictly between the extreme tips of the wings if the vertical component of the Earth's magnetic field at that location is $1.75 \times 10^{-5} \text{ T}$? [PYQ 2018] | **[Chapter 6: Electromagnetic Induction | Confidence: High]**

Q20. Briefly write two important, real-world practical applications each for (i) X-rays, and (ii) Microwaves. [PYQ 2017] | **[Chapter 8: Electromagnetic Waves | Confidence: Very High]**

Q21. The threshold work function of pure caesium metal is precisely 2.14 eV . Calculate the exact threshold frequency for the emission of photoelectrons from this metal. (Given $h = 6.63 \times 10^{-34} \text{ J s}$, $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$). [PYQ 2018] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section C: 3-Mark Questions (Short Answer)

Q22. Derive the mathematical expression for the torque experienced by a short electric dipole of dipole moment $\vec{p}$ placed in a uniform external electric field $\vec{E}$. Additionally, calculate the work done required to slowly rotate this dipole from its position of stable equilibrium to unstable equilibrium. [PYQ 2014, 2020] | **[Chapter 1: Electric Charges & Fields | Confidence: Very High]**

Q23. Three ideal capacitors of capacitances C_1 , C_2 , and C_3 are connected strictly in parallel across a voltage source V . Derive the formal expression for the equivalent capacitance of this combination. Also, show

mathematically that the total electrostatic energy stored in this parallel combination is exactly equal to the sum of the energies stored in each individual capacitor. [PYQ 2019] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: High]**

Q24. State Ampere's circuital law. Using this law, systematically derive the expression for the magnetic field strictly inside a long, tightly wound, straight solenoid carrying a steady current I and having n turns per unit length. [PYQ 2014, 2016] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q25. Define the following fundamental terms pertaining to the magnetic properties of materials: (i) Magnetic permeability (μ), (ii) Magnetic susceptibility (χ_m), and (iii) Magnetization ($\vec{M}$). Write the standard mathematical relation connecting relative magnetic permeability (μ_r) and magnetic susceptibility (χ_m). [PYQ 2018, 2020] | **[Chapter 5: Magnetism & Matter | Confidence: Very High]**

Q26. An AC voltage given by $V = V_0 \sin \omega t$ is applied across a series combination of a pure inductor L and a pure resistor R . Using a properly labelled phasor diagram, derive the mathematical expression for the net impedance (Z) of the circuit and the phase angle (ϕ) between the alternating current and the applied voltage. [Practice / PYQ 2023] | **[Chapter 7: Alternating Current | Confidence: Very High]**

Q27. Draw a fully labelled geometric ray diagram illustrating the formation of a real image of a point object placed on the principal axis of a concave spherical mirror. Using this diagram, rigorously derive the standard mirror formula: $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$. [PYQ 2018] | **[Chapter 9: Ray Optics | Confidence: High]**

Q28. Based on Thomas Young's double-slit experiment (YDSE), logically explain the specific effect on the interference fringes observed on the screen when: (i) the observation screen is moved physically closer to the plane of the slits. (ii) the monochromatic light source is completely replaced by a source of white light. (iii) the physical width of the single source slit illuminating the two slits is substantially increased. Justify your answers briefly. [PYQ 2022] | **[Chapter 10: Wave Optics | Confidence: Very High]**

Section D: 4-Mark Questions (Case-Study Based)

Q29. Case Study: Kirchhoff's Laws for Complex Circuits In complex electrical circuits where Ohm's law alone is insufficient or cumbersome to apply, Kirchhoff's two rules are utilized. Kirchhoff's first law (the Junction Rule) states that the algebraic sum of currents meeting at any electrical junction is strictly zero. Kirchhoff's second law (the Loop Rule) dictates that the algebraic sum of changes in potential around any closed circuit loop is exactly zero. These rules strictly obey fundamental laws of nature and provide a robust method to determine the unknown currents and voltages in multi-loop networks. (A) On which two distinct fundamental conservation principles are Kirchhoff's first and second rules based, respectively? (1 Mark) (B) While applying the loop rule, if you traverse a closed loop in the exact direction of the assumed current through a resistor, what is the accepted sign convention for the potential difference across it? (1 Mark) (C) In a given circuit, two non-ideal cells of emfs 1.5 V and 2.0 V , with internal resistances 1Ω and 2Ω respectively, are connected strictly in parallel (positive terminals joined together). This entire combination is then connected across an external load resistance of 5Ω . Using Kirchhoff's rules (or equivalent formulas), determine the exact terminal potential difference across the 5Ω external resistor. (2 Marks) [PYQ 2016, 2021] | **[Chapter 3: Current Electricity | Confidence: Very High]**

Q30. Case Study: The Photoelectric Effect The photoelectric effect is the spontaneous emission of electrons when electromagnetic radiation, such as UV light, hits a metallic material. The electrons emitted in this manner are called photoelectrons. The minimum frequency of light required to successfully emit electrons is defined as the threshold frequency. The maximum kinetic energy of the emitted photoelectrons increases linearly with the frequency of the incident light, while the saturation photoelectric current is directly proportional to the intensity of the incident light, provided the frequency is strictly above the threshold. (A) How does the stopping potential of the emitted photoelectrons depend on the physical intensity of the incident light? (1 Mark) (B) What is the exact mathematical effect on the maximum kinetic energy of photoelectrons if the frequency of incident light is suddenly doubled (assuming it remains above threshold)? (1 Mark) (C) Monochromatic light of

frequency $7.21 \times 10^{14} \text{ Hz}$ is incident on a metal surface. Electrons with a maximum speed of $6.0 \times 10^5 \text{ m/s}$ are successfully ejected from the surface. Calculate the threshold frequency for the photoemission of electrons for this specific metal. (Given $h = 6.64 \times 10^{-34} \text{ J s}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$). (2 Marks) [PYQ 2018, 2023] | **[Chapter 11: Dual Nature of Radiation & Matter | Confidence: Very High]**

Section E: 5-Mark Questions (Long Answer)

Q31. (A) Define the term 'Electrostatic Potential' at a point. Is it a scalar or a vector quantity? (B) Derive the general mathematical expression for the electrostatic potential V at a point P situated at a distance r from an isolated point charge $+q$. (C) Three point charges $+q$, $+2q$, and $-3q$ are placed respectively at the three vertices of an equilateral triangle of side length L . Find the exact total work done required by an external agent to completely separate these three charges to an infinite distance from each other. [PYQ 2020 / Practice] | **[Chapter 2: Electrostatic Potential & Capacitance | Confidence: Very High]**

Q32. (A) State the Biot-Savart law expressing the magnetic field due to a current element in vector form. (B) Apply the Biot-Savart law to rigorously derive the mathematical expression for the magnetic field intensity (B) at a point situated on the axis of a circular current-carrying loop of radius R at a distance x from its centre. (C) Two identical circular conducting coils, P and Q , each of radius R , carrying steady currents of 1 A and $\sqrt{3} \text{ A}$ respectively, are placed concentrically and perpendicular to each other. Coil P lies in the XY plane and coil Q lies in the YZ plane. Find the exact magnitude and the direction (angle) of the net magnetic field produced exactly at the common centre of the coils. [PYQ 2016, 2017] | **[Chapter 4: Moving Charges & Magnetism | Confidence: Very High]**

Q33. (A) What is meant by the phenomenon of 'Total Internal Reflection' in optics? (B) State the two essential optical conditions required for total internal reflection to occur successfully. (C) Derive the mathematical relation between the critical angle (θ_c) and the refractive index (μ) of the optically denser medium with respect to the rarer medium. (D) A point source of monochromatic light is placed at the bottom of a transparent liquid of refractive index $\mu = 5/3$ filled in a tank to a depth of $h = 2 \text{ m}$. Calculate the exact area of the circular surface on the top of the liquid through which the light from the source can emerge out into the air. [PYQ 2019, 2022] | **[Chapter 9: Ray Optics | Confidence: Very High]**

Answer Key

Q1: (b) $q/8\epsilon_0$. A point charge at a corner is shared equally by 8 adjacent identical cubes. The flux through one single cube is $1/8^{\text{th}}$ of the total flux q/ϵ_0 . **Q2:** (b) $Q/4\pi\epsilon_0 R$ and Zero. Inside a charged conductor, the electric field is strictly zero, and the potential is uniform everywhere and equal to its value on the surface. **Q3:** (b) decreases strictly with an increase in temperature. As temperature rises, thermal agitation increases, causing free electrons to collide more frequently with lattice ions, dropping the relaxation time. **Q4:** (b) $2r$. The radius of a circular path in a magnetic field is given by $r = mv/qB$. Radius is directly proportional to velocity. If v doubles, r exactly doubles. **Q5:** (c) Ferromagnetic. Ferromagnetic materials have relative permeabilities (μ_r) in the hundreds or thousands, making their absolute permeability μ maximum. **Q6:** (b) rate of change of magnetic flux linked with the circuit. ($\epsilon = -d\Phi/dt$). **Q7:** (b) become exactly half. Inductive reactance $X_L = \omega L = 2\pi fL$. Current $I = V/X_L$. If frequency f is doubled, X_L doubles, so current strictly halves. **Q8:** (b) $2 \times 10^{-7} \text{ T}$. Amplitude ratio $E_0/B_0 = c \implies B_0 = E_0/c = 60/(3 \times 10^8) = 20 \times 10^{-8} = 2 \times 10^{-7} \text{ T}$. **Q9:** (a) f . Cutting a lens along the principal axis splits it into two lenses with the exact same radii of curvature for both surfaces. By the Lens Maker's formula, the focal length f strictly remains unchanged. **Q10:** (b) shift downwards. The central bright fringe forms where the optical path difference is zero. If the source shifts upwards, the lower ray path length decreases, so the zero-path-difference point shifts geometrically downwards. **Q11:** (b) the frequency of the incident light and the nature of the material. ($K_{\text{max}} = h\nu - \Phi$).

Q12: (a) Electron. By de-Broglie equation $\lambda = h/mv$. For identical velocities, $\lambda \propto 1/m$. The electron has the lowest mass, hence the longest wavelength.

Q13: (a) Both A and R are true and R is the correct explanation of A. **Q14:** (b) Both A and R are true but R is NOT the correct explanation of A. The correct mathematical reason is $1/R_{eq} = 1/R_1 + 1/R_2 + \dots$, ensuring R_{eq} is smaller than the smallest resistor. **Q15:** (c) A is true but R is false. Biot-Savart law states $dB \propto 1/r^2$ (square of the distance). The $1/r$ dependence for an infinite wire comes after integration. **Q16:** (a) Both A and R are true and R is the correct explanation of A. By Lens Maker's formula, if $\mu_{liquid} > \mu_{lens}$, the factor $(\mu_{lens}/\mu_{liquid} - 1)$ becomes negative, reversing its optical nature.

Q17: Energy $U = \frac{1}{2} CV^2 = \frac{1}{2} \times (12 \times 10^{-12} \text{ F}) \times (50 \text{ V})^2 = 6 \times 10^{-12} \times 2500 = 1.5 \times 10^{-8} \text{ J}$. **Q18:** Drift velocity (v_d) is the average uniform velocity acquired by free electrons in a conductor drifting towards the positive terminal under an applied electric field. Relation: $I = neAv_d$. **Q19:** Velocity $v = 1080 \text{ km/h} = 1080 \times \frac{5}{18} = 300 \text{ m/s}$. Motional emf $e = B_vlv = (1.75 \times 10^{-5} \text{ T}) \times (40 \text{ m}) \times (300 \text{ m/s}) = 1.75 \times 10^{-5} \times 12000 = 0.21 \text{ V}$. **Q20:** (i) X-rays: Medical radiography (detecting bone fractures) and studying crystal structures. (ii) Microwaves: Radar systems for aircraft navigation and microwave ovens for heating food. **Q21:** Work function $\Phi = h\nu_0 \implies \nu_0 = \Phi/h$. Converting eV to Joules: $\Phi = 2.14 \times 1.6 \times 10^{-19} = 3.424 \times 10^{-19} \text{ J}$. Threshold frequency $\nu_0 = (3.424 \times 10^{-19})/(6.63 \times 10^{-34}) \approx 5.16 \times 10^{14} \text{ Hz}$.

Q22: Consider a dipole with charges $+q$ and $-q$ separated by $2a$ in uniform field E . Forces are $+qE$ and $-qE$, forming a couple. Torque

$\tau = \text{Force} \times \text{Perpendicular Distance} = qE \times 2a \sin \theta = (q \times 2a)E \sin \theta = pE \sin \theta$. Vectorially,

$\vec{\tau} = \vec{p} \times \vec{E}$. Stable equilibrium is at $\theta = 0^\circ$; unstable is at $\theta = 180^\circ$. Work done

$W = \int_0^{180^\circ} pE \sin \theta d\theta = pE[-\cos \theta]_0^{180^\circ} = pE[-(-1) - (-1)] = 2pE$. **Q23:** In parallel, voltage V is same. Total charge $Q = Q_1 + Q_2 + Q_3 = C_1V + C_2V + C_3V$. Also $Q = C_{eq}V$. Thus $C_{eq} = C_1 + C_2 + C_3$. Total energy $U_{tot} = \frac{1}{2} C_{eq} V^2 = \frac{1}{2} (C_1 + C_2 + C_3) V^2 = \frac{1}{2} C_1 V^2 + \frac{1}{2} C_2 V^2 + \frac{1}{2} C_3 V^2 = U_1 + U_2 + U_3$. **Q24:**

Ampere's Law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$. Consider a rectangular Amperian loop ABCD of length l partially inside the solenoid. Only the segment inside (length l) contributes to the integral: $\int_A^B B dl \cos(0^\circ) = Bl$. The segments perpendicular to B yield zero, and the outside segment yields zero ($B \approx 0$). Total turns in loop = nl . Enclosed current $I_{enc} = (nl)I$. Thus, $Bl = \mu_0(nl)I \implies B = \mu_0 nI$. **Q25:** (i) Magnetic permeability (μ): The ability of a material to permit the passage of magnetic field lines through it. (ii) Susceptibility (χ_m): A measure of how easily a substance can be magnetized in an external field ($\vec{M} = \chi_m \vec{H}$). (iii) Magnetization ($\vec{M}$): The net magnetic dipole moment developed per unit volume of the material. Relation: $\mu_r = 1 + \chi_m$. **Q26:** Draw phasor diagram: V_R is in phase with current I ; V_L leads I by 90° . Total voltage V is the vector sum:

$V = \sqrt{V_R^2 + V_L^2} = \sqrt{(IR)^2 + (IX_L)^2} = I\sqrt{R^2 + X_L^2}$. Impedance $Z = V/I = \sqrt{R^2 + X_L^2}$. From the phasor triangle, phase angle ϕ is given by $\tan \phi = V_L/V_R = IX_L/IR = X_L/R$. The current lags the voltage by angle ϕ . **Q27:** Draw a concave mirror. Object AB beyond C. Real, inverted image A'B' between C and F. Draw ray parallel to axis passing through F, and a ray striking pole P reflecting symmetrically. Triangles $\triangle A'B'P$ and $\triangle ABP$ are similar: $A'B'/AB = B'P/BP$. Also $\triangle A'B'F$ and $\triangle MPF$ (where M is point of incidence of parallel ray) are similar: $A'B'/MP = B'F/PF$. Since $MP = AB$, $B'P/BP = B'F/PF$. Using sign convention: $BP = -u$, $B'P = -v$, $PF = -f$. $B'F = -v - (-f) = -v + f$. Substitute:

$-v/-u = (-v + f)/-f \implies v/u = (v - f)/f = v/f - 1$. Divide by v : $1/u = 1/f - 1/v \implies 1/f = 1/v + 1/u$. **Q28:** (i) Fringe width $\beta = \lambda D/d$. Decreasing screen distance D strictly decreases the fringe width β . (ii) The central fringe remains perfectly white (all colors overlap with zero path difference), but the secondary fringes become colored, with violet closest to the center and red furthest, quickly overlapping into general illumination. (iii) A wide source slit acts as multiple independent narrow slits side-by-side. Their individual interference patterns overlap incoherently, causing the bright and dark fringes to wash out, severely reducing pattern sharpness and ultimately making fringes disappear.

Q29: (A) First rule (Junction) is based on the Law of Conservation of Charge. Second rule (Loop) is based on the Law of Conservation of Energy. (B) The potential difference is taken as negative ($-IR$), representing a potential

drop. (C) First, find the equivalent emf and internal resistance of the parallel cells:

$$E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} = \frac{(1.5 \times 2) + (2.0 \times 1)}{1 + 2} = \frac{3.0 + 2.0}{3} = \frac{5.0}{3} \text{ V. Equivalent internal resistance}$$

$$r_{eq} = \frac{r_1 r_2}{r_1 + r_2} = \frac{1 \times 2}{1 + 2} = 2/3 \Omega. \text{ Current in circuit } I = \frac{E_{eq}}{R_{ext} + r_{eq}} = \frac{5/3}{5 + 2/3} = \frac{5/3}{17/3} = \frac{5}{17} \text{ A. Terminal voltage}$$

across the 5Ω load is $V = IR_{ext} = (5/17) \times 5 = 25/17 \approx 1.47 \text{ V}$. **Q30:** (A) The stopping potential is completely independent of the physical intensity of the incident light. (B) $K_{max} = h\nu - \Phi$. If ν is doubled (2ν), $K'_{max} = 2h\nu - \Phi$. This is strictly more than double the original kinetic energy ($2(h\nu - \Phi) = 2h\nu - 2\Phi$). (C) Kinetic energy

$$K_{max} = \frac{1}{2} m_e v^2 = \frac{1}{2} \times 9.1 \times 10^{-31} \times (6.0 \times 10^5)^2 = 0.5 \times 9.1 \times 10^{-31} \times 36 \times 10^{10} = 1.638 \times 10^{-19} \text{ J}$$

. By Einstein's equation: $h\nu = \Phi + K_{max} \Rightarrow h\nu = h\nu_0 + K_{max} \Rightarrow \nu_0 = \nu - (K_{max}/h)$.

$$\nu_0 = 7.21 \times 10^{14} - \frac{1.638 \times 10^{-19}}{6.64 \times 10^{-34}} \approx 7.21 \times 10^{14} - 2.47 \times 10^{14} = 4.74 \times 10^{14} \text{ Hz.}$$

Q31: (A) Electrostatic potential at a point is defined as the total work done by an external force in bringing a unit positive test charge from infinity to that point without acceleration. It is strictly a scalar quantity. (B) Consider a test charge q_0 at distance x from $+q$. Force $F = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{x^2}$. Work done to move it dx against force is $dW = -Fdx$. Total work from ∞ to r is $W = \int_{\infty}^r -\frac{1}{4\pi\epsilon_0} \frac{qq_0}{x^2} dx = -\frac{qq_0}{4\pi\epsilon_0} [-1/x]_{\infty}^r = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r}$. Potential $V = W/q_0 = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$. (C) The work done by an external agent to separate the charges to infinity is equal to the negative of the initial electrostatic potential energy of the system. Initial

$$U = \frac{1}{4\pi\epsilon_0} \left[\frac{(q)(2q)}{L} + \frac{(2q)(-3q)}{L} + \frac{(-3q)(q)}{L} \right] = \frac{k}{L} [2q^2 - 6q^2 - 3q^2] = -\frac{7kq^2}{L}. \text{ Separation Work}$$

$W = 0 - U = 0 - (-7kq^2/L) = +\frac{7q^2}{4\pi\epsilon_0 L}$. **Q32:** (A) $d\vec{B} = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \vec{r}}{r^3}$. (B) Current loop in y-z plane. Take an element $d\vec{l}$ at the top. The distance to point P on x-axis is $r = \sqrt{R^2 + x^2}$. $dB = \frac{\mu_0 I dl}{4\pi(R^2 + x^2)}$. Resolving dB , the perpendicular components strictly cancel out. The axial components ($dB \sin \theta$, where $\sin \theta = R/r$) add up. Integrating $d\vec{l}$ gives $2\pi R$. Net field $B = \frac{\mu_0 I}{4\pi(R^2 + x^2)} \frac{R}{\sqrt{R^2 + x^2}} (2\pi R) = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$. (C) The magnetic field exactly

at the centre ($x = 0$) is $B = \mu_0 I / 2R$. Field of coil P (in XY plane) points along the Z-axis: $B_P = \frac{\mu_0(1)}{2R} = \frac{\mu_0}{2R}$. Field of coil Q (in YZ plane) points along the X-axis: $B_Q = \frac{\mu_0(\sqrt{3})}{2R} = \frac{\mu_0\sqrt{3}}{2R}$. Since fields are perpendicular, net magnitude $B_{net} = \sqrt{B_P^2 + B_Q^2} = \frac{\mu_0}{2R} \sqrt{1^2 + (\sqrt{3})^2} = \frac{\mu_0}{2R} \sqrt{4} = \frac{2\mu_0}{2R} = \frac{\mu_0}{R}$. Direction makes an angle θ with the X-axis: $\tan \theta = B_P/B_Q = 1/\sqrt{3} \Rightarrow \theta = 30^\circ$ with the X-axis (or 60° with the Z-axis). **Q33:** (A)

When a ray of light travelling from an optically denser medium to a rarer medium strikes the interface at an angle strictly greater than the critical angle, it is completely reflected back into the denser medium. (B) Conditions: 1. Light must strictly travel from a denser to a rarer medium. 2. Angle of incidence must be strictly greater than the critical angle ($i > \theta_c$). (C) At the critical angle ($i = \theta_c$), the angle of refraction $r = 90^\circ$. By Snell's Law at the interface: $\mu_d \sin \theta_c = \mu_r \sin 90^\circ = \mu_r \times 1$. Thus, $\sin \theta_c = \mu_r / \mu_d$. If the rarer medium is air ($\mu_r = 1$), then $\sin \theta_c = 1/\mu$. (D) The light emerges through a cone whose semi-vertical angle is exactly θ_c . Radius of the illuminated circle $r = h \tan \theta_c$. From $\sin \theta_c = 1/\mu$, we mathematically get $\tan \theta_c = \frac{1}{\sqrt{\mu^2 - 1}}$.

Area $A = \pi r^2 = \pi h^2 \tan^2 \theta_c = \frac{\pi h^2}{\mu^2 - 1}$. Given $\mu = 5/3 \Rightarrow \mu^2 - 1 = 25/9 - 1 = 16/9$. Area

$$A = \frac{\pi(2)^2}{16/9} = \frac{4\pi}{16/9} = \frac{36\pi}{16} = \frac{9\pi}{4} \approx 7.07 \text{ m}^2.$$